$\boldsymbol{Jamalpur\ Science\ \&\ Technology\ University}$

$\boldsymbol{Department\ of\ Electrical\ and\ Electronic\ Engineering}$

$\boldsymbol{B.Sc.(Eng.)1st\ year\ 2nd\ semester\ final\ examination - 2023}$

$\boldsymbol{Course\ code: MATH\ 1211}$

$\boldsymbol{Course\ title: Linear\ Algebra, Coordinate\ Geometry\ and\ Vector\ Analysis}$

Full marks: 70 Time: 3 hours

(**N.B.** Answer any five questions. The figures in the right margin indicate full marks.)

1. (a) Define vector space with example. 3

   (b) Let U and W be the subspace of $\mathbb{R}^4$ generated by $\{(1,1,0,-1),(1,2,3,0),(2,3,3,-1)\}$ and $\{(1,2,2,-2),(2,3,2,-3),(1,3,4,-3)\}$ respectively. Find the basis and dimension of $U+W$. 5

   (c) Define direct sum. In $\mathbb{R}^3$, let $U=\{(a,b,0): a,b\in\mathbb{R}\}$ and $W=\{(0,0,c): c\in\mathbb{R}\}$. Show that $\mathbb{R}^3 = U \oplus W$. 1+5

2. (a) What do you mean by linearly independent and dependent vectors. If $\boldsymbol{u},\boldsymbol{v}$ and $\boldsymbol{w}$ are linearly independent vectors, then show that the vectors $\boldsymbol{u}+\boldsymbol{v}, \boldsymbol{u}-\boldsymbol{v}$ and $\boldsymbol{u}-2\boldsymbol{v}+\boldsymbol{w}$ are linearly independent. 2+3

   (b) Let $T: U \to V$ be a linear transformation. Then show that (i) the image of T is a subspace of V and (ii) the kernel of T is a subspace of U. 2+2

   (c) Let $T: \mathbb{R}^3 \to \mathbb{R}^3$ be a linear operator defined by $T(x,y,z) = (2x, 4x-y, 2x+3y-z)$. Then (i) Show that T is non-singular and (ii) Find a formula for T^{-1}. 2+3

3. (a) Define a matrix representation of a linear operator. Let T be a linear operator in $\mathbb{R}^3$ defined by $T(x,y,z) = (2x+z, x-4y, 3x)$, find the matrix representation of T relative to the basis $\{\boldsymbol{e_1}=(1,1,1), \boldsymbol{e_2}=(1,1,0), \boldsymbol{e_3}=(1,0,0)\}$. 1+4

   (b) Find $T(a,b)$ where $T: \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(3,1)=(2,-4)$ and $T(1,1)=(0,2)$. 5

   (c) Determine the inverse of the following matrix by using row canonical form: 4

$$A = \begin{bmatrix} 3 & 4 & -1 \\ 1 & 0 & 3 \\ 2 & 5 & -4 \end{bmatrix}$$

4. (a) Define rank of a matrix. Find the rank of the matrix $A = \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 2 & 3 & 7 & 5 \\ 2 & 5 & 11 & 6 \end{bmatrix}$ by reducing it to the normal form. 1+4

   (b) Find all eigenvalues and the corresponding eigenvectors of the matrix $A = \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix}$. 5

   (c) Test whether the following system of linear equation is consistent or not. If it is consistent then solve it. 4

$$x + 2y - z = 2$$
$$2x + y + z = 1$$
$$x + 5 - 4z = 5$$

5. (a) Determine the transformation equation of $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ when the origin remain unchanged and the direction of axes is turned through an angle $tan^{-1}\left(\frac{-4}{3}\right)$. 5

(b) Reduce the conic $8x^2 + 4xy + 5y^2 - 16x - 14y + 13 = 0$ to the standard form and identify it. 4

(c) Define direction cosines and direction ratios of a line. If l, m, n be the direction cosines of any line AB, then prove that $l^2 + m^2 + n^2 = 1$. 2+3

6. (a) Prove that the angle between two diagonals of a cube is $cos^{-1}\left(\frac{1}{3}\right)$. 4

(b) A variable plane is at constant distance p from the origin and meets the axes in A, B and C. Show that the locus of the centroid of the tetrahedron $OABC$ is $x^{-2} + y^{-2} + z^{-2} = 16p^{-2}$ 5

(c) A particle moves along the curve $x = 2t^2$, $y = t^2 - 4t$, $z = 3t - 5$, where t is the time. Find the components of its velocity and acceleration at time $t = 1$ in the direction $\hat{\imath} - 3\hat{\jmath} + 2\hat{k}$. 5

7. (a) Define gradient of a function. If $\varphi = ln|\vec{r}|$, where $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$, then find $\nabla\varphi$. 1+3

(b) What do you mean by curl and divergence? If $\vec{A} = xz^3\hat{\imath} - 2x^2yz\hat{\jmath} + 2yz^4\hat{k}$, then find curl $\vec{A}$. 2+3

(c) State Green's theorem in the plane and verify the theorem in the plane for $\oint_C (xy + y^2)\,dx + x^2dy$ where C is the closed curve of the region bounded by $y = x$ and $y = x^2$. 1+4